

# Seminario

# Multimedia Forensics

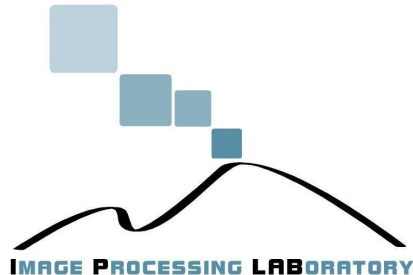
# DCT nella Image Forensics

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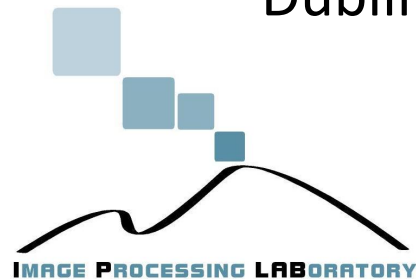
DMI – University of Catania



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DFF – ACM Multimedia  
Dublin 2025



Visiting Research Period  
Vigo 2026

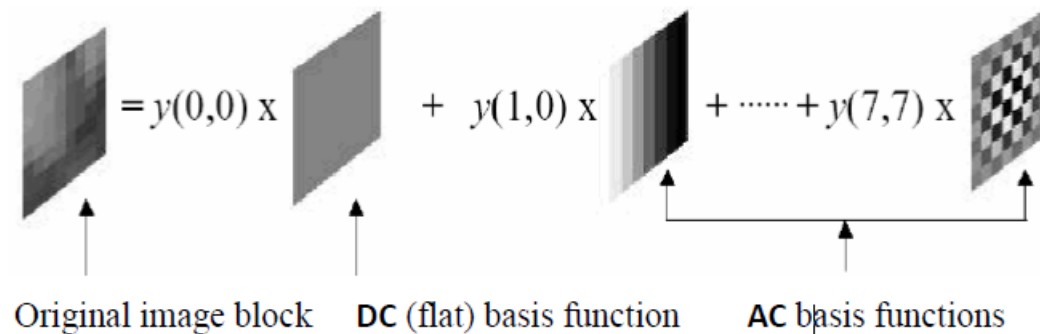


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# Image Representation with DCT

An image can be analyzed either in the spatial or frequency field

## Discrete Cosine Transform



$$C(u) = \alpha(u) \sum_{x=0}^{N-1} f(x) \cos \left[ \frac{\pi(2x+1)u}{2N} \right]$$

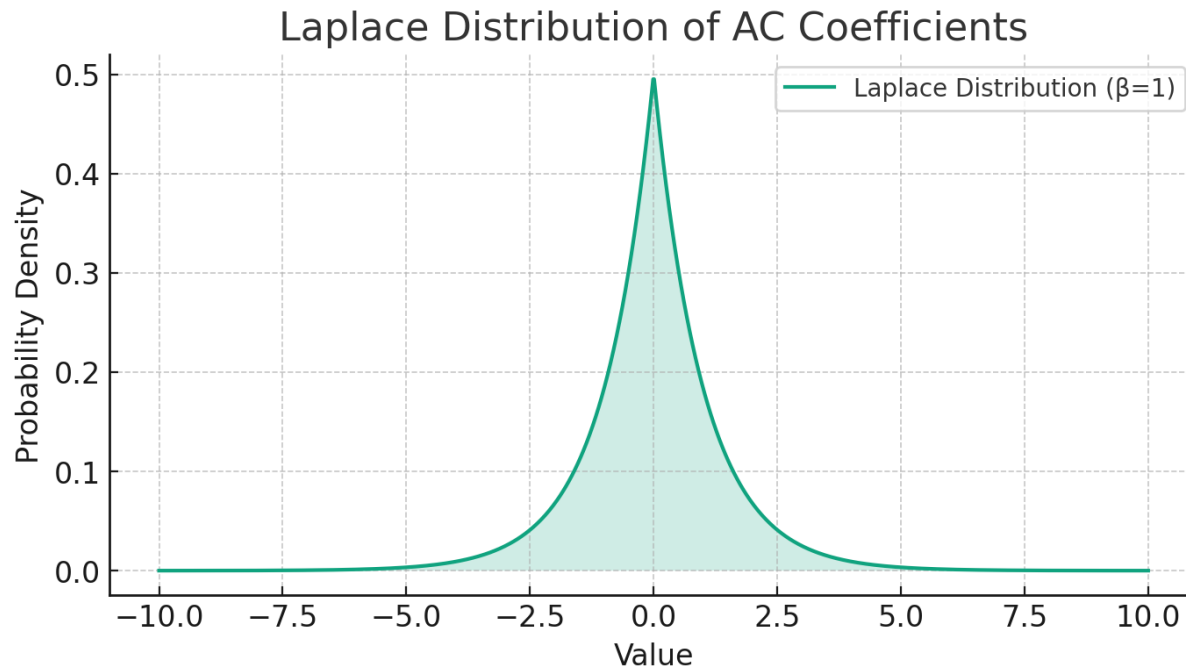
Onedimensional Discrete Cosine Transform (DCT)

$$DCT_{2D}(I) = DCT(DCT(I_T)_T)$$

2-dimensional DCT

# AC Coefficient

The statistical distribution of the AC DCT coefficients can be approximated by a **Laplacian** distribution.  
(Lam and Goodman, 2000)



$\beta$  is a scale parameter,  
crucial for determining  
the distribution's spread

$\mu$  is the average,  
indicates the peak of the  
laplacian

# PROBLEM 1

Field

- Image forensics

Target

- Identifying cropped images

Why is it important to know if an image has been cropped?

- To understand if the image has been tampered - Authenticity and Integrity of the image



# Why Cropping Detection?



# Why Cropping Detection?

| LINEE DERIVATE DAL Q.E.G. DI PIANO |                                               |                |           |           |
|------------------------------------|-----------------------------------------------|----------------|-----------|-----------|
| LINEA                              | UTENZA                                        | DOSSALE (mm)   | TIPO CAVO | TIPO CAVO |
| 8                                  | Alimentazione Nuovo Q.E. MECCANICHELETTICISTI | 3x70-3x35-3x25 | PGT       | PGT       |

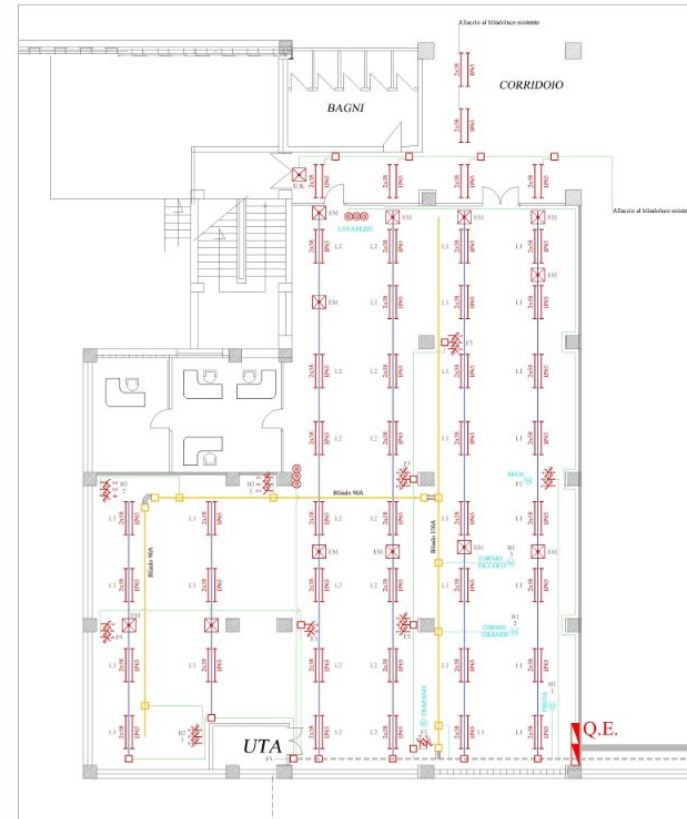
| LINEE DERIVATE DAL NUOVO QUADRO MECCANICO - ELETTRICISTI |                            |              |           |           |
|----------------------------------------------------------|----------------------------|--------------|-----------|-----------|
| LINEA                                                    | UTENZA                     | DOSSALE (mm) | TIPO CAVO | TIPO CAVO |
| L1                                                       | Linea circuito 1           | 302,5        | PGT ERM   | 2x1,5+T   |
| L2                                                       | Linea circuito 2           | 302,5        | PGT ERM   | 2x1,5+T   |
| L3                                                       | Linea circuito 3           | 302,5        | PGT ERM   | 2x1,5+T   |
| L4                                                       | Alimentazione di emergenza | 302,5        | PGT ERM   | 2x1,5     |
| F1                                                       | PM BLINDO 150A             | 3x240-3x120  | PGT ERM   | ---       |
| F2                                                       | Press. area 1 (Segg)       | 4x4-7x4      | MOT V.M.  | 4x4-7x4   |
| F3                                                       | Press. area 2 (Cappano)    | 4x4-7x4      | MOT V.M.  | 4x4-7x4   |
| F4                                                       | Press. area 3              | 4x4-7x4      | MOT V.M.  | 4x4-7x4   |
| F5                                                       | Q.E. LTA                   | 108          | PGT ERM   | ---       |

| LINEE DERIVATE DAL BLINDO 150A ALLE MACCHINE |                |      |         |     |
|----------------------------------------------|----------------|------|---------|-----|
| B1-1                                         | FRIGIDA        | SC30 | PGT ERM | --- |
| B1-2                                         | TORNIO GRANDE  | SC30 | PGT ERM | --- |
| B1-3                                         | TORNIO PICCOLO | SC30 | PGT ERM | --- |

| LINEE DERIVATE DAL BLINDO 150A ALLE PRESE INTERBLOCATE |                            |         |          |     |
|--------------------------------------------------------|----------------------------|---------|----------|-----|
| B2-1                                                   | Gruppo prese interbloccate | 4x4-7x4 | MOT V.M. | --- |
| B2-2                                                   | Gruppo prese interbloccate | 4x4-7x4 | MOT V.M. | --- |
| B2-3                                                   | Gruppo prese interbloccate | 4x4-7x4 | MOT V.M. | --- |

# Why Cropping Detection?





# RAISE Dataset (.TIF images)



4288x2848



Duc-Tien Dang-Nguyen,  
Pasquini, C., Conotter, V., and  
Boato, G. (2015).  
RAISE: a Raw Images Dataset  
for Digital Image Forensics.

8156 Raw Images

Split 80% Train – 20% Test



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# Centrally cropped images (aligned scenario)



2848x2848

# Y Channel and 5 Resolutions



2048x2048



1024x1024



512x512



256x256

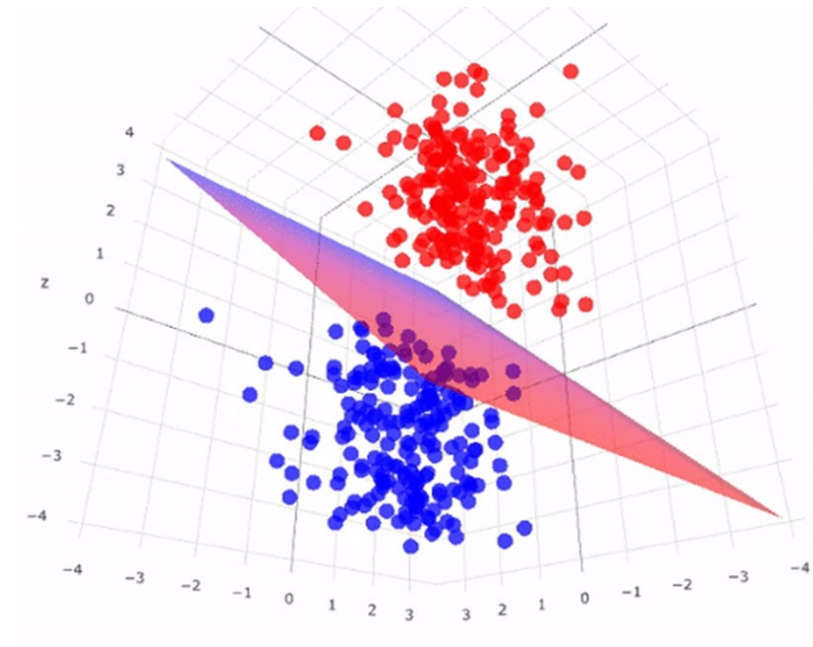


128x128

# Feature extraction and Training of SVM

Train SVM which classifies  
the image in 5 resolution  
classes using  $\beta$

Feature extraction: the 63  $\beta$  are  
computed and collected in a csv file.  
The result is a matrix where each row is  
an image and each column is a beta  
coefficient (so 63 columns).

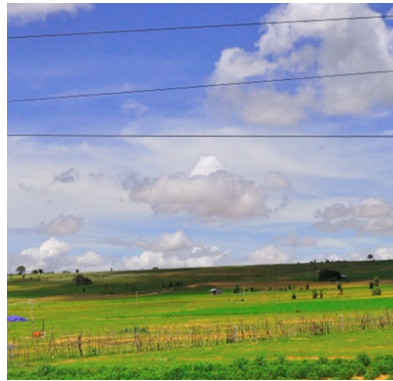




# Cropping Detection



2048x2048



512x512

Resolution Classification  
using AC coefficient  
features ( $\beta$ )

Classified Resolution is:  
 $2048 \times 2048 \neq$  input image resolution  
( $512 \times 512$ )

Image is **Cropped!**



**IMAGE PROCESSING LABORATORY**



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# Test Result

| <div> <div>Predicted</div> <div>Real</div> </div> | 2048 × 2048 | 1024 × 1024 | 512 × 512 | 256 × 256 | 128 × 128 |
|---------------------------------------------------|-------------|-------------|-----------|-----------|-----------|
|                                                   | 2048 × 2048 | 1024 × 1024 | 512 × 512 | 256 × 256 | 128 × 128 |
| 2048 × 2048                                       | 98.43%      | 1.02%       | 0.14%     | 0.00%     | 0.41%     |
| 1024 × 1024                                       | 1.49%       | 87.79%      | 6.72%     | 1.76%     | 2.24%     |
| 512 × 512                                         | 1.02%       | 13.09%      | 64.76%    | 17.86%    | 3.27%     |
| 256 × 256                                         | 0.43%       | 6.77%       | 27.93%    | 53.64%    | 11.23%    |
| 128 × 128                                         | 0.35%       | 4.01%       | 3.18%     | 15.57%    | 76.9%     |

Confusion Matrix of the SVM Classifier

# Test Result

| Predicted<br>resolution<br>Cropping size | 2048 × 2048 | 1024 × 1024 | 512 × 512 | 256 × 256 | 128 × 128 | % crop detection |
|------------------------------------------|-------------|-------------|-----------|-----------|-----------|------------------|
|                                          |             |             |           |           |           |                  |
| 1024 × 1024                              | 99%         | 1%          | 0%        | 0%        | 0%        | 99%              |
| 512 × 512                                | 89.5%       | 4%          | 1%        | 0.5%      | 5%        | 93.5%            |
| 256 × 256                                | 82.5%       | 4.5%        | 2%        | 0.5%      | 10.5%     | 89%              |
| 128 × 128                                | 76%         | 3%          | 1.5%      | 3%        | 16.5%     | 83.5%            |

Crop detection accuracy and classification results of images cropped from 2048×2048 resolution with cropping sizes equal to classification categories



# Test Result

| Predicted<br>resolution<br>Cropping size | 2048 × 2048 | 1024 × 1024 | 512 × 512 | 256 × 256 | 128 × 128 | % crop detection |
|------------------------------------------|-------------|-------------|-----------|-----------|-----------|------------------|
|                                          |             |             |           |           |           |                  |
| 1536 × 1536                              | 100%        | 0%          | 0%        | 0%        | 0%        | 100%             |
| 750 × 750                                | 96%         | 2%          | 0%        | 0%        | 2%        | 98%              |
| 350 × 350                                | 85.5%       | 4.5%        | 1.5%      | 1%        | 7.5%      | 91.5%            |

Crop detection accuracy and classification results of images cropped from 2048×2048 resolution with cropping sizes different w.r.t classification categories

# PROBLEM 2

Field

- Image forensics

Target

- Accurate Estimate of Geometrical Transformation in the JPEG Images

Why is it important to know if an image has been cropped?

- Existing JPEG forensic methods (DQD, FQE, Forgery Detection) dramatically lose performance when geometric transformations are applied

# THE HIDDEN PROBLEMS

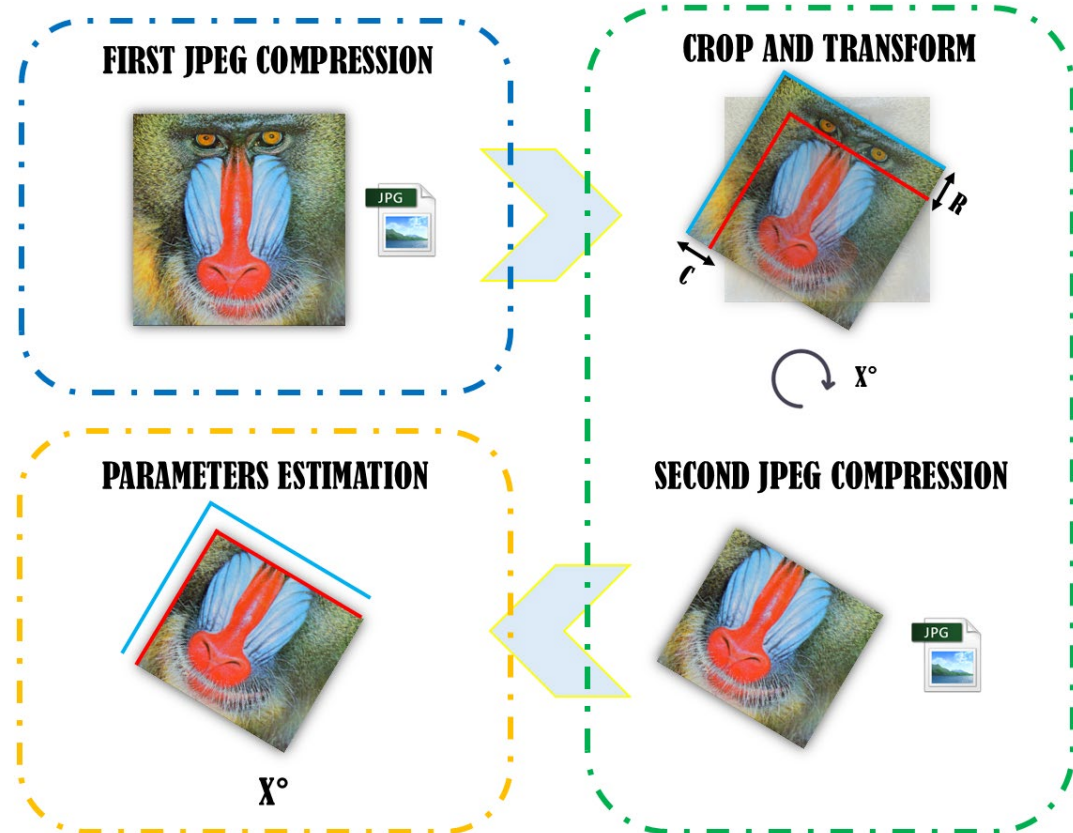
- Resize, Rotation, etc interpolates pixels
- The 8×8 grid is misaligned (Not aligned scenario)



SOTA methods are not designed for this

# THE SOLUTION

Accurately estimate the parameters of the applied geometric transformation, then apply the inverse transformation and restore the effectiveness of existing methods.



# Impact of Geometric Transformations

How much do existing methods degrade performance?

Tested methods:

[Bianchi et al.](#)

Image forgery localization via block-grained analysis of JPEG artifacts

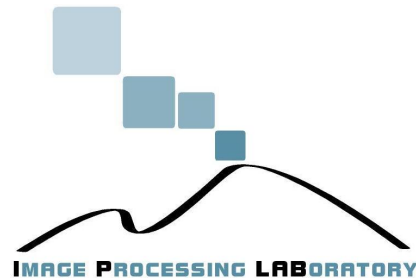
[Battiato et al.](#)

Cnn-based first quantization estimation of double compressed jpeg images

Experimental setup:

1000 RAISE images (512×512) |  $QF_1 \in \{55, 60, 65, 70, 75, 80, 85\}$  |  $QF_2 = 90$

Scaling: -20 to +20 pixels (1px steps) | Rotation: -10° to +10° (0.5° steps)



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# Forgery Localization via JPEG Artifacts

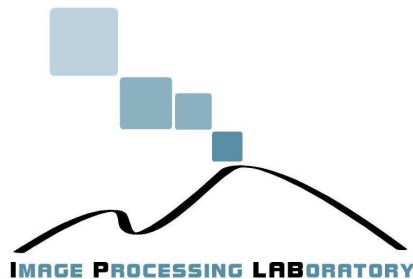
*Bianchi & Piva, IEEE TIFS 2012*

## What it does

- Detects double JPEG compression artifacts
- Localizes forged regions at 8×8 block level
- Works with both Aligned (A-DJPG) and Non-Aligned (NA-DJPG) scenarios
- Outputs likelihood map indicating tampering probability

## Why it matters for our work

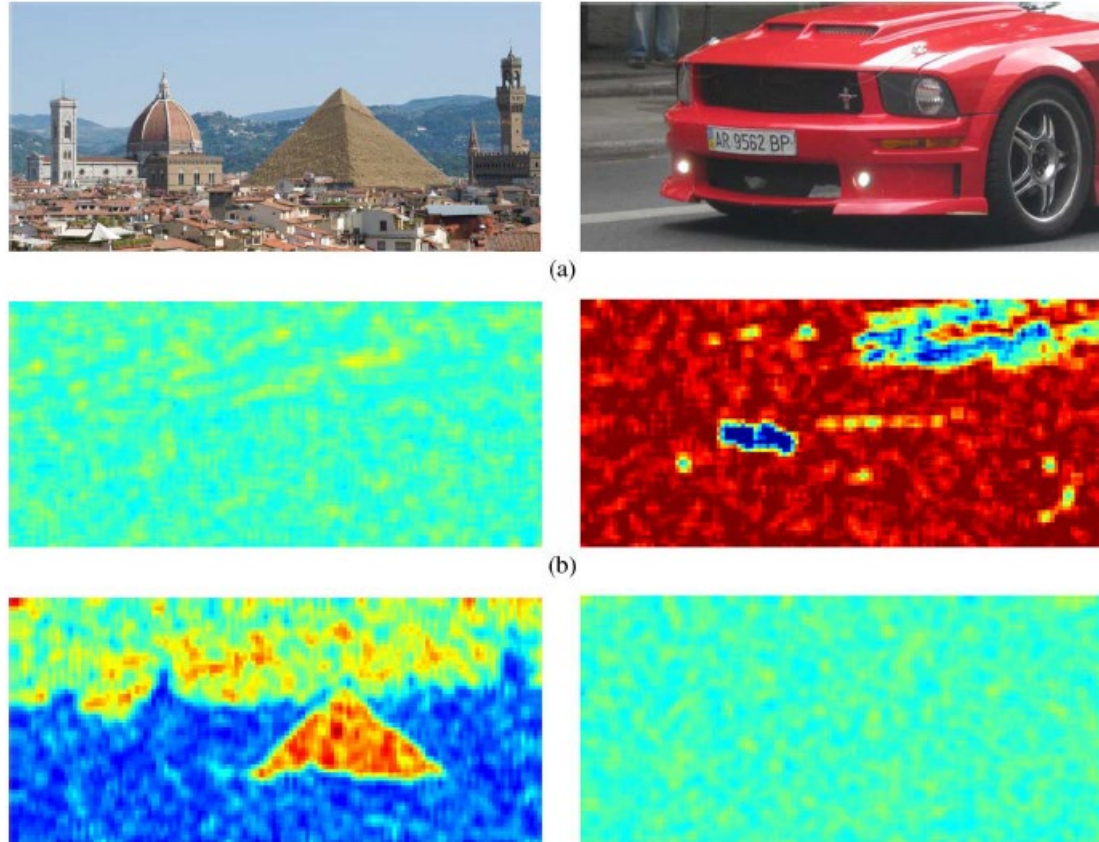
- Foundational method for JPEG forensics
- Statistical model of DCT coefficient distributions
- EM algorithm for  $Q_1$  estimation
- Widely cited baseline for FQE methods



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# Forgery Localization via JPEG Artifacts

*Bianchi & Piva, IEEE TIFS 2012*



# CNN-based First Quantization Estimation

*Battiato et al., JVCIR 2022*

## What it does

- Estimates  $Q_1$  matrix from double JPEG images
- CNN-based approach on DCT histograms
- Works with custom quantization matrices
- Flexible input patch sizes (64-512 px)

## Key innovations

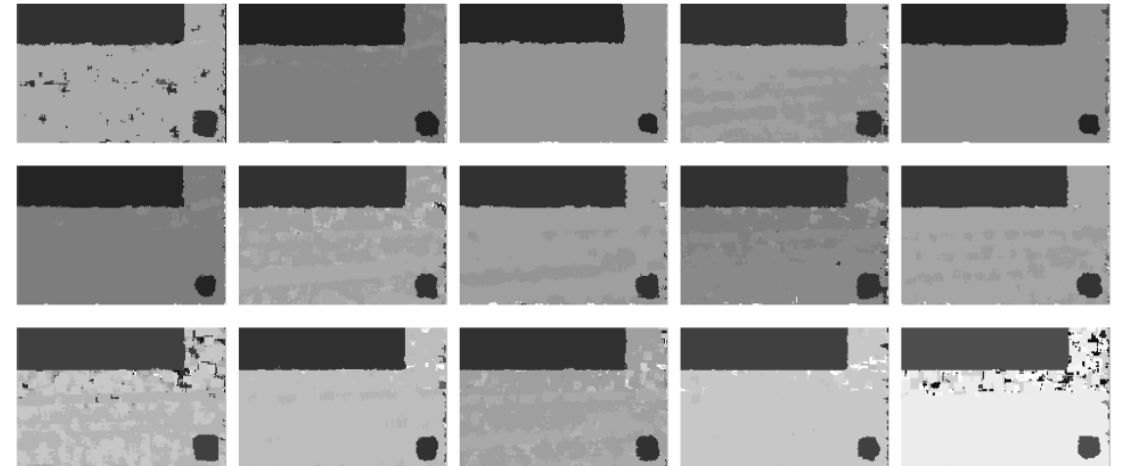
- Ensemble of CNNs (one per  $q_2$  value)
- Regularization using neighbor constraints
- Outperforms statistical methods
- Tampering localization via sliding window



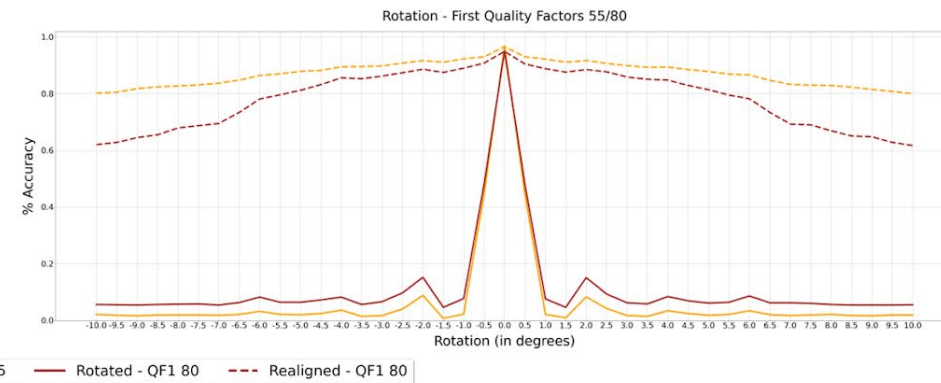
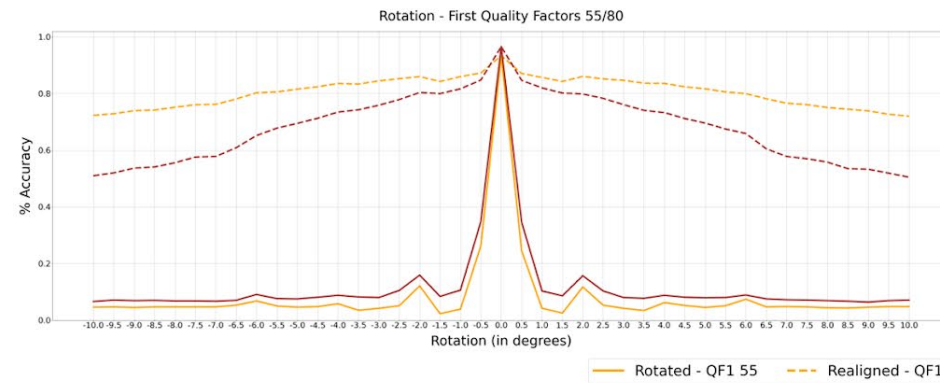
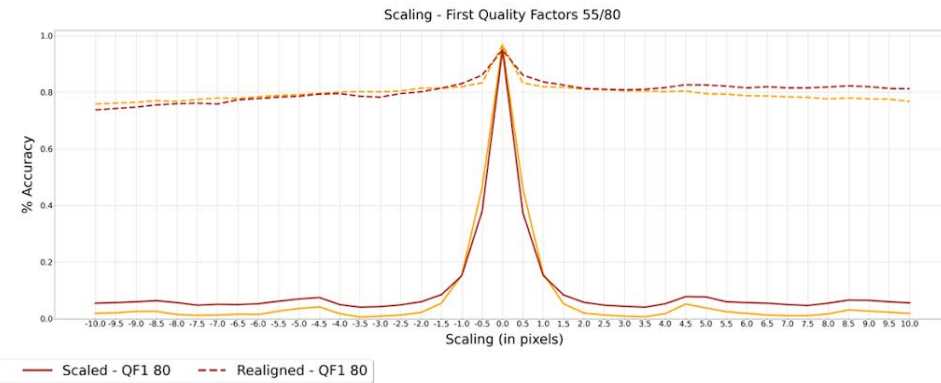
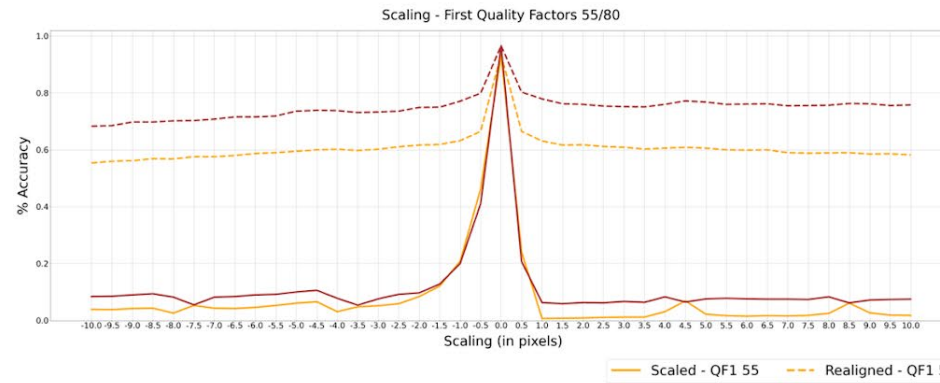
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# CNN-based First Quantization Estimation

*Battiato et al., JVCIR 2022*



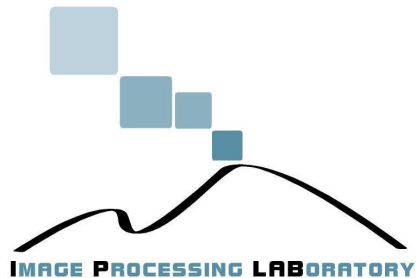
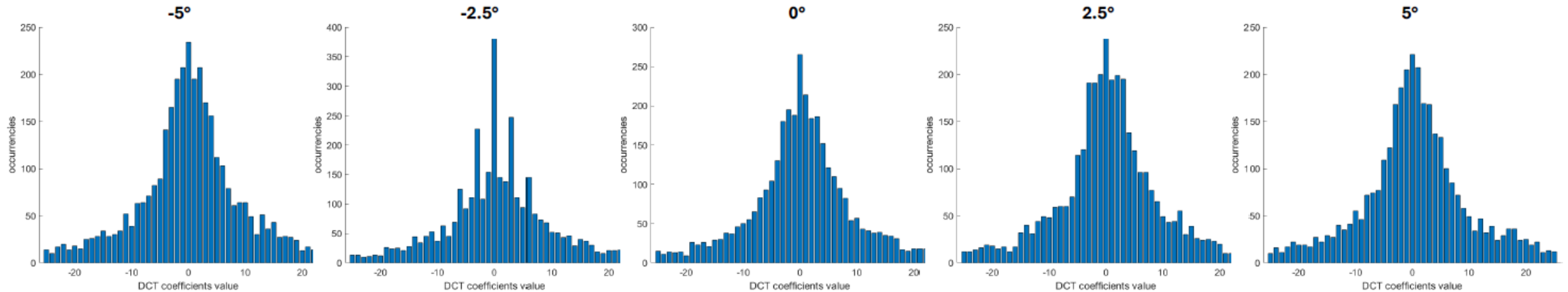
# Impact of Geometric Transformations





# Insight

JPEG traces reappear in the DCT histogram when we apply the correct inverse transformation. This distinctive pattern allows us to identify the correct parameters among all candidates.



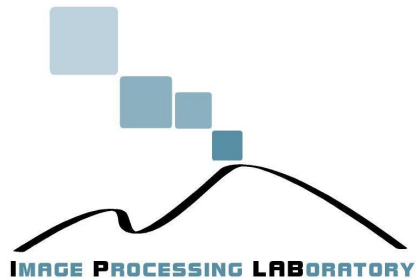
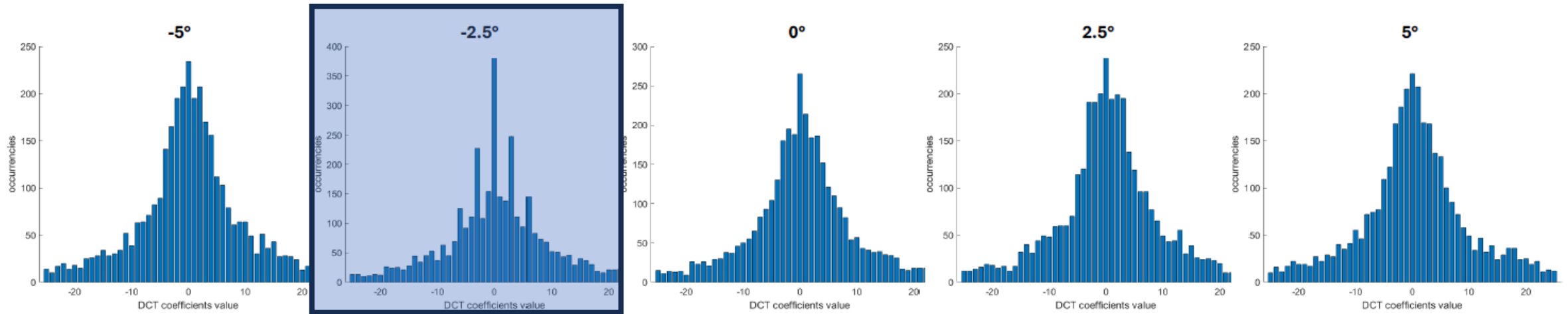
JPG image with original rotation of +2.5°  
Various inverse transformations applied



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# Insight

Only when we apply the correct inverse transformation ( $-2.5^\circ$ ) does the DCT histogram show the characteristic "comb" pattern with regular peaks due to JPEG quantization. The other histograms show the classic Laplacian distribution without obvious artifacts.



JPG image with original rotation of  $+2.5^\circ$   
Various inverse transformations applied



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# Proposed Method: Overview

1

## Input

DJPEG image  
with unknown  $T_{\text{true}}$

2

## Candidates

Generate set of  
inverse transforms  $T_k$

3

## Apply + Compress

For each  $T_k$ :  
apply + 3rd  
compression

4

## Compute Distance

Compare DCT  
histograms  
with reference

5

## Estimate

$T^* = \operatorname{argmax} D_{tk}$   
 $\hat{T} = T^{*-1}$

# Proposed Method: Pseudo Algorithm

INPUT: Double JPEG image  $I$  with unknown transformation  $T_{\text{true}}$

**1 STEP 1: Candidate Generation**  
Generate candidate inverse transformations  $T_k$   
Rotation:  $\theta \in [-\theta_{\text{max}}, +\theta_{\text{max}}]$ , step  $\theta_{\text{step}}$   
Scaling:  $s \in [s_{\text{min}}, s_{\text{max}}]$

**2 STEP 2: Apply & Compress**  
For each candidate  $T_k$ :  
• Apply inverse transformation:  $I_k = T_k(I)$   
• Apply 3rd JPEG compression:  $I'_k = \text{JPEG}(I_k, \text{QF}_2)$

**3 STEP 3: Build Reference & Compute Distance**  
For each  $T_k$  and crop offset  $c \in \{0, \dots, 63\}$ :  
• Extract DCT histogram:  $h^f_{\{T_k, c\}}$   
• Build reference:  $h^f_{\{\text{ref}\}} = \text{mean over 64 crops}$   
• Compute:  $D_{\{T_k, c\}} = \sum_f \text{SED}(h^f_{\{T_k, c\}}, h^f_{\{\text{ref}\}})$

**4 STEP 4: L1 Normalization**

Normalize distance across crop dimension:  
 $\hat{D}_{\{T_k\}} = \max_c \{D_{\{T_k, c\}}\} / \sum_c D_{\{T_k, c\}}$   
Penalizes uniform distributions (wrong  $T_k$ )

**5 STEP 5: Select Best Candidate**

Find transformation that MAXIMIZES distance:  
 $T^* = \text{argmax}_{\{T_k\}} \hat{D}_{\{T_k\}}$   
Highest distance = JPEG traces visible = correct

**OUTPUT: Estimated transformation  $\hat{T} = (T^*)^{-1}$**



*Two-Step Strategy: Step 1-3 run with coarse parameters (256 bins, 16 crops) → select top 25 candidates → Step 1-5 run with fine parameters (1024 bins, 64 crops) for final estimation*



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# Building the Reference Histogram

We cannot directly compare histograms of different transformations (they might have different trends).

## **Solution:**

For each candidate transformation, we build a reference histogram with limited JPEG artifacts.

64 different crops ( $N_r, N_c \in \{0, \dots, 7\}$ )  $\rightarrow$  compression  $\rightarrow$  average of histograms



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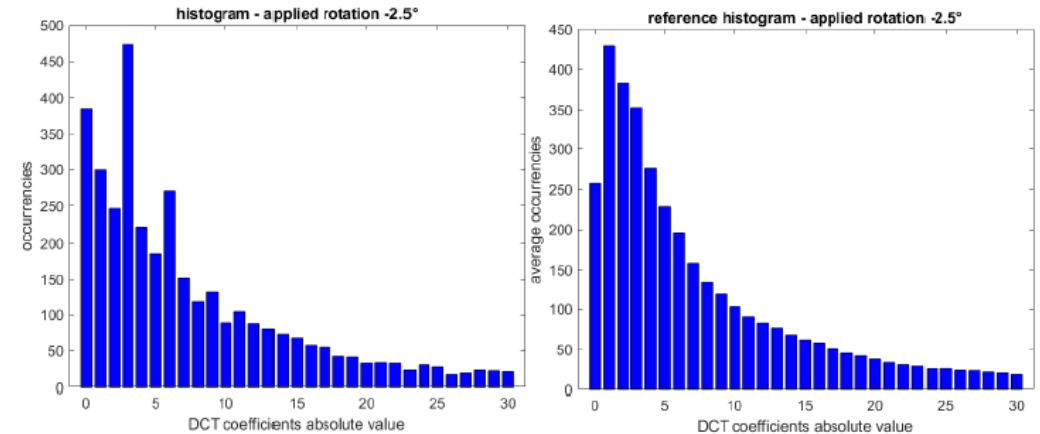
# Building the Reference Histogram

**(Correct -2.5°):**

Histogram shows "comb" pattern (JPEG signature)

Reference is smooth (no artifacts)

**Distance = HIGH ✓**

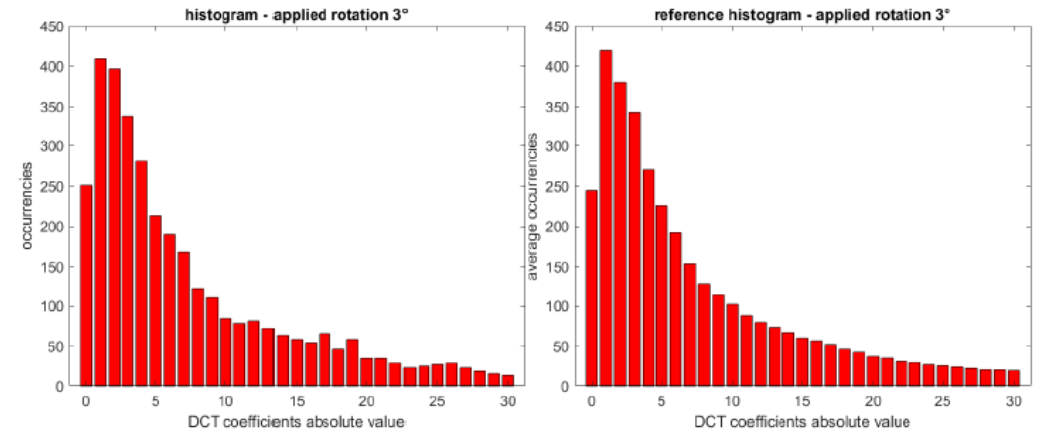


**(Wrong 3°):**

Histogram is smooth (no signature)

Reference is similar

**Distance = LOW ✗**



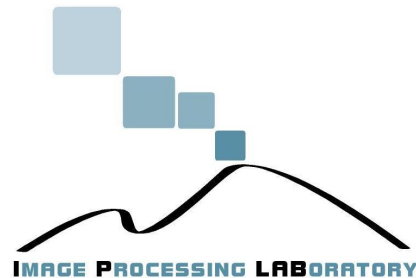
# Two Step Strategy

## Step 1: Candidate Selection

- FAST but less accurate version
- 256 bins for histograms
- 16 crops out of 64 possible
- Explores wide range of parameters
- Selects Ncand = 25 peaks

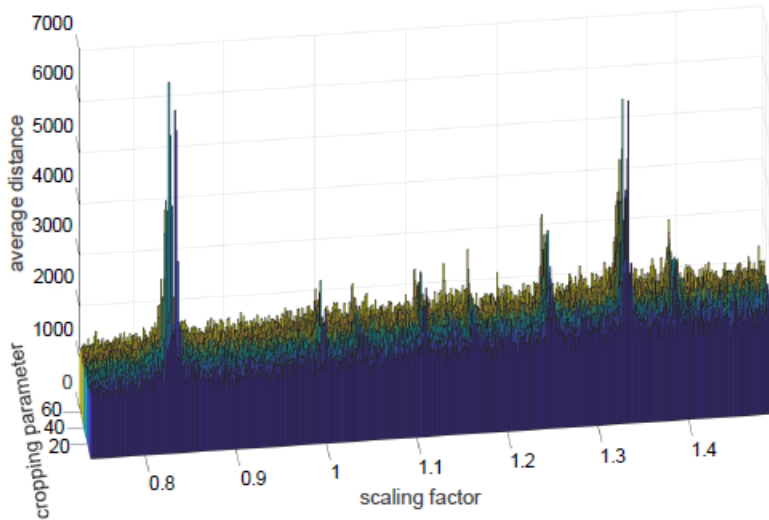
## Step 2: Final Estimation

- ACCURATE version
- 1024 bins for histograms
- All 64 crops
- Expansion to neighboring values
- L1 normalization for crop



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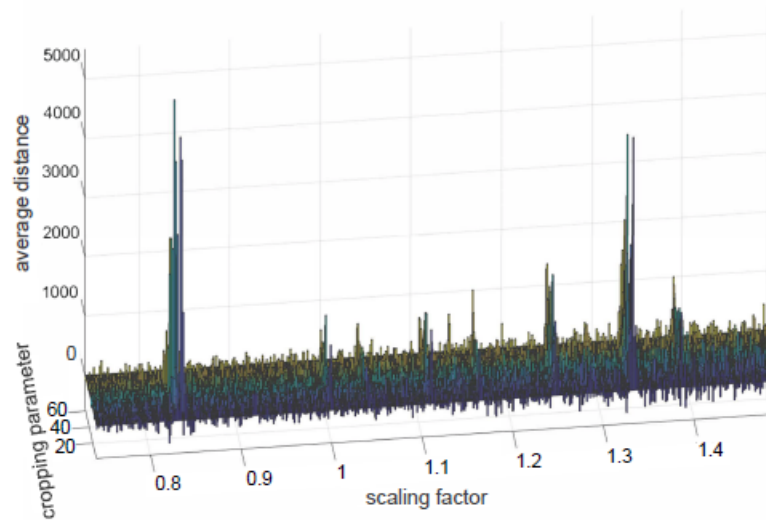
# Two Step Strategy



(a)

Raw Distance

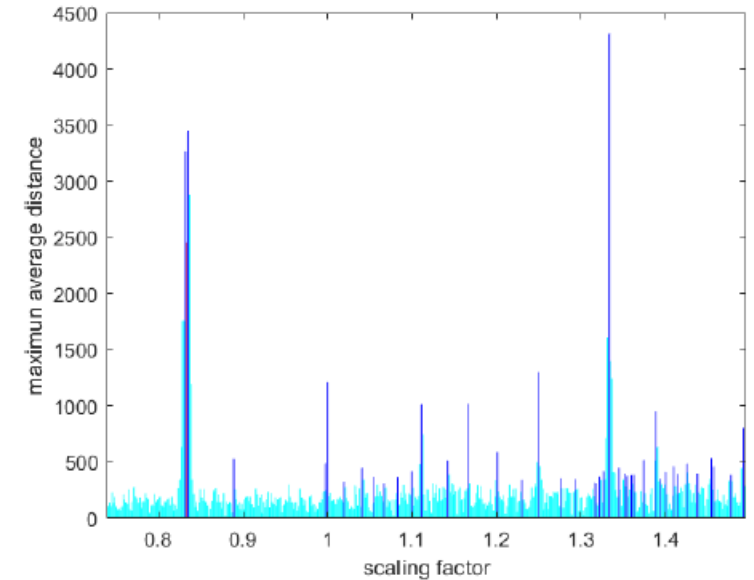
Distance function over all candidates



(b)

After Filtering

Median filter removes trends



(c)

Peak Selection

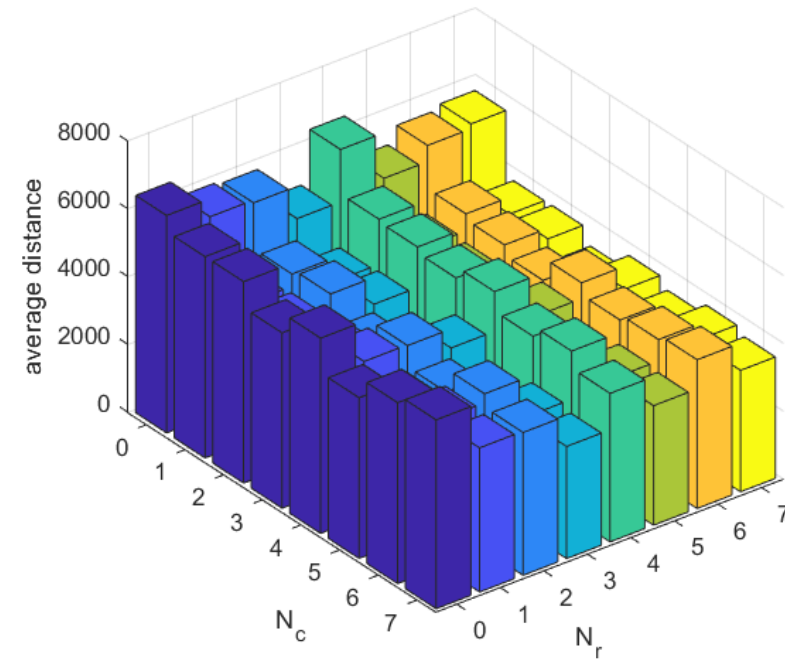
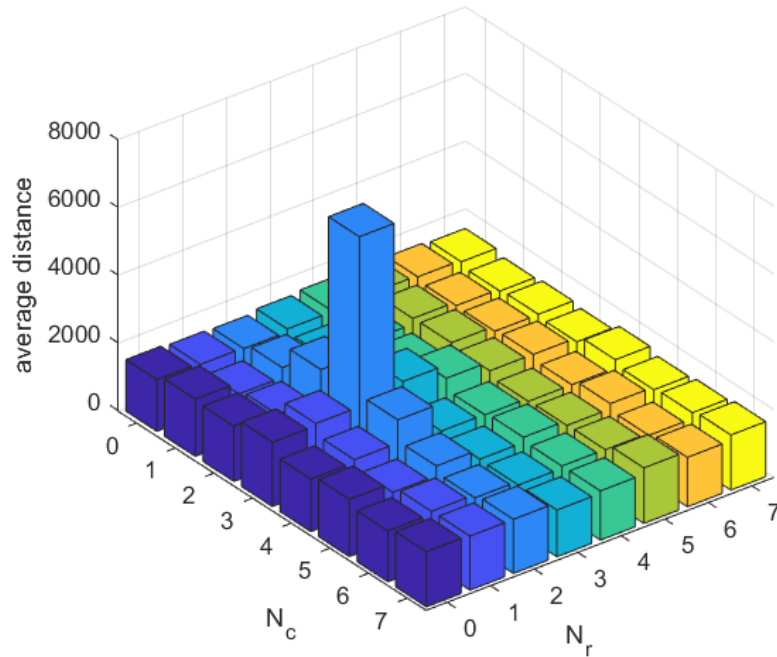
Max projection → top 25

Peaks = high distance from reference = likely correct transformation



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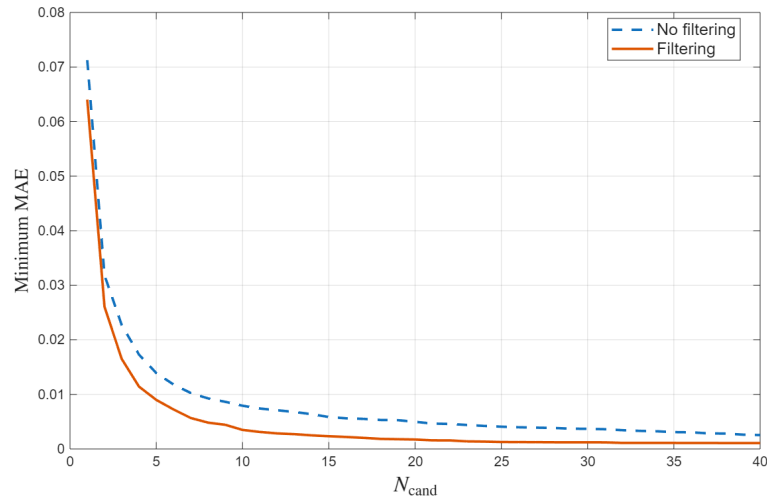
# L1 Normalization



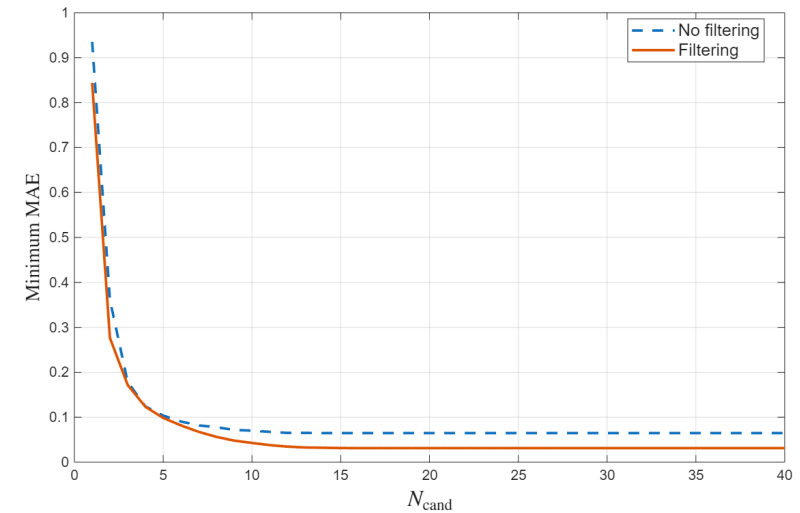
The  $8 \times 8$  distribution ( $N_r \times N_c$ ) shows a clear peak in the correct case  $\rightarrow$  L1 normalization penalizes uniform distributions

# Validation: Number of Candidates

*How many candidates to select in Step 1?*



(a) Resize scenario

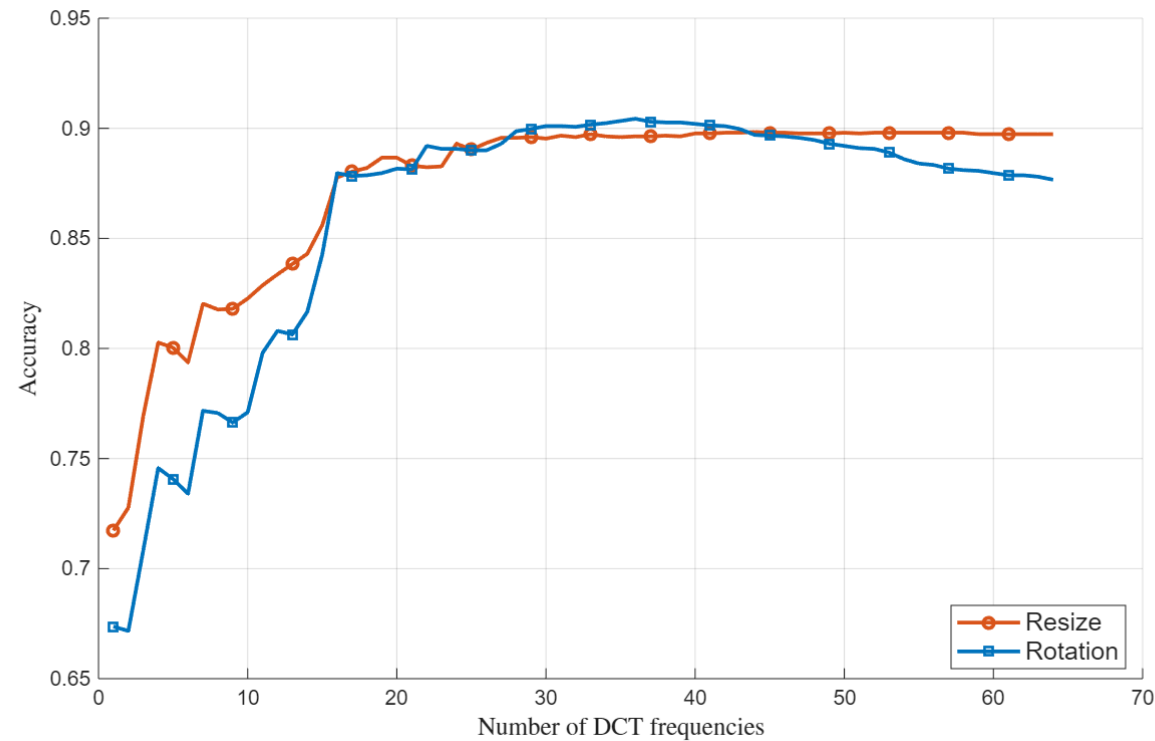


(a) Rotation scenario

**Minimum MAE stabilizes a  $N_{\text{cand}} \approx 25$**   
**Filtering always improves performance**

# Validation: Number of Candidates

*How many DCT frequencies to include in the summation?  $\rightarrow f_{max} = 40$*

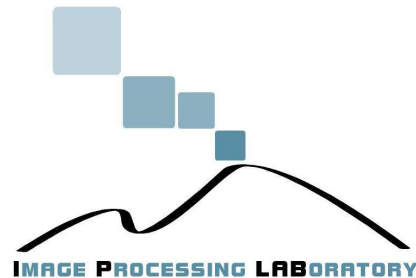




# Test Result

## Resolution Variation

| Transf.  | Resolution | Avg.  | QF1=55 | QF1=70 | QF1=85 |
|----------|------------|-------|--------|--------|--------|
| Rotation | 256×256    | 83.5% | 94.5%  | 94.9%  | 26.4%  |
|          | 512×512    | 91.3% | 96.7%  | 99.3%  | 49.6%  |
|          | 1024×1024  | 96.6% | 100%   | 99.8%  | 77.3%  |
| Resize   | 256×256    | 85.7% | 93.1%  | 93.0%  | 50.2%  |
|          | 512×512    | 91.9% | 97.6%  | 97.4%  | 67.4%  |
|          | 1024×1024  | 93.7% | 97.4%  | 99.5%  | 70.4%  |



Accuracy increases with resolution (more information available), even at 256×256, we achieve >83% (challenging NA-DJPEG scenario). Low QF1 (more artifacts) → better results; High QF1 → more difficult scenario



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# Test Result

## Shift Estimation

$P(T \mid \text{Correct Crop})$

$\approx 99.8\%$

If we correctly estimate the crop,  
the transformation is almost always correct

$P(T \mid \text{Wrong Crop})$

$\approx 1-5\%$

If we get the crop wrong,  
the transformation is almost always wrong

The bottleneck is Shift estimation. Once the offset ( $N_r$ ,  $N_c$ ) is correctly identified, the transformation parameter (angle or scale) is estimated with very high precision. This justifies the L1 normalization strategy for crop.

# Test Result

## QF2 Variation

| Transf.  | QF2 | Avg.  | QF1=55 | QF1=70 | QF1=85 |
|----------|-----|-------|--------|--------|--------|
| Rotation | 80  | 68.9% | 96.9%  | 96.2%  | 16.0%  |
|          | 85  | 85.2% | 97.4%  | 98.1%  | 19.5%  |
|          | 90  | 91.3% | 96.7%  | 99.3%  | 49.6%  |
| Resize   | 80  | 52.3% | 90.0%  | 49.8%  | 14.6%  |
|          | 85  | 69.7% | 95.9%  | 80.5%  | 28.5%  |
|          | 90  | 91.9% | 97.6%  | 97.4%  | 67.4%  |

# Test Result

## Dataset Generalization

| RAISE                                            |       | BOSSBase                                         |       | Dresden                                    |       |
|--------------------------------------------------|-------|--------------------------------------------------|-------|--------------------------------------------|-------|
| Dataset for tuning<br>RAW images, various scenes |       | 10K grayscale 512x512<br>Uncompressed PGM format |       | 70+ camera models<br>High resolution, JPEG |       |
| Rotation                                         | 91.3% | Rotation                                         | 95.3% | Rotation                                   | 96.5% |
| Resize                                           | 91.9% | Resize                                           | 92.6% | Resize                                     | 95.4% |

# Test Result

## Transformation Generalization: Shear

### What is Shear?

Linear transformation that 'tilts' the image by shifting rows/columns proportionally.

Tested angles: 82.5° - 97.5°  
(step 2.5°)

### Average Accuracy

# 97%

across all QF1 and shear angles

| QF1  | 55    | 60    | 65    | 70    | 75    | 80   | 85    |
|------|-------|-------|-------|-------|-------|------|-------|
| Acc. | 97.2% | 95.5% | 98.0% | 98.0% | 99.5% | 100% | 93.8% |



Method is transformation-agnostic



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# Test Result

## SOTA Comparison: Scaling Factor Estimation

| Method       | Type          | 256 X 256    | 512 X 512    | 1024 X 1024  |
|--------------|---------------|--------------|--------------|--------------|
| Bianchi [26] | Guided        | 86.2%        | 92.4%        | 95.6%        |
| Sahu [32]    | Guided        | 51.7%        | 67.0%        | 59.0%        |
| Billa [39]   | Guided        | -            | 24.1%        | 65.5%        |
| <b>Ours</b>  | <b>Guided</b> | <b>91.3%</b> | <b>96.7%</b> | <b>97.7%</b> |
| Bianchi [26] | Blind         | 17.1%        | 38.1%        | 42.7%        |
| Sahu [32]    | Blind         | 0.6%         | 5.4%         | 7.3%         |
| Choi [42]    | Blind         | -            | 20.9%        | 19.5%        |
| <b>Ours</b>  | <b>Blind</b>  | <b>85.0%</b> | <b>91.9%</b> | <b>92.5%</b> |



# Test Result

## SOTA Comparison: Rotation Angle Estimation

| Method             | 256 X 256    | 512 X 512    | 1024 X 1024  | MAE 512      |
|--------------------|--------------|--------------|--------------|--------------|
| Chen [28]          | 34.0%        | 44.2%        | 69.8%        | 3.26°        |
| Vazquez-Padín [27] | 1.3%         | 2.2%         | 28.9%        | 7.73°        |
| Choi [42]          | 32.3%        | 66.8%        | 69.1%        | 0.49°        |
| <b>Ours</b>        | <b>83.6%</b> | <b>91.6%</b> | <b>95.5%</b> | <b>0.38°</b> |

We outperform all methods in both Guided and Blind modes, consistent performance between guided and blind (our method does not need it)), MAE significantly lower than all competitors

# PROBLEM 3

## What is JPEG AI?

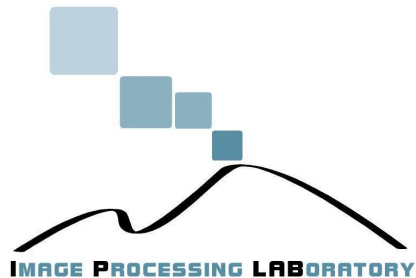
- New JPEG standard (2023) using neural networks
- End-to-end learned compression
- Latent space representation instead of DCT
- Designed for both compression AND analysis

## Our Contribution (DFF '25)

- Retrieval directly on compressed latents
- No decoding required → efficiency gains
- Deepfake source attribution via k-NN
- ~77% Top-1 accuracy on 7 generators

Key concept: JPEG AI latents preserve information?

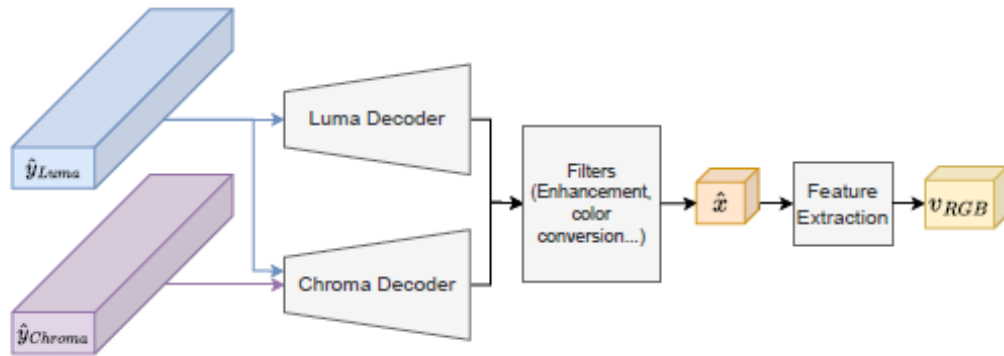
We can investigate directly in latents → privacy-preserving forensics.



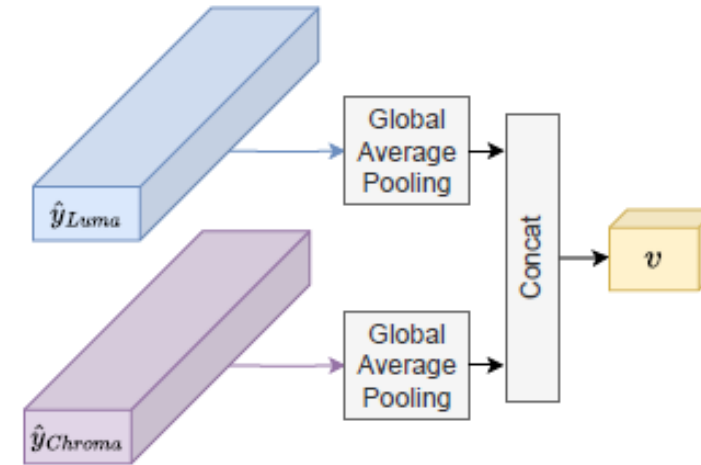
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# Our Pipeline

Traditional pipeline



Ours pipeline



Key concept: JPEG AI latents preserve information?

We can investigate directly in latents → privacy-preserving forensics.

# Solved Tasks



(a) Query



(b) ✓



(c) ✗



(d) ✗



(e) ✓



(f) ✓



(g) ✗



(h) ✓



(i) ✓



(j) ✓



(k) ✓

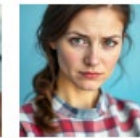
Content Based Image Retrieval



(a) Query (Deep AI)



(b) ✓✓



(c) ✓✓



(d) ✓✗



(e) ✓✗



(f) ✓✓



(g) ✓✗



(h) ✓✗



(i) ✓✗



(j) ✓✗



(k) ✓✓

Deepfake Source Attribution

# PROBLEM 4

## What is HEIF?

- High Efficiency Image File Format
- Default format on iPhone since iOS 11 (2017)
- Uses HEVC (H.265) video codec for images
- ~50% smaller than JPEG at same quality

## Why it matters for forensics

- Billions of iPhones worldwide
- Different compression artifacts than JPEG
- Limited forensic research so far
- Critical gap in current methods

### Questions:

1. Can existing JPEG forensic methods be adapted to HEIF?
2. What artifacts does HEVC compression leave in images?
3. How do double HEIF compressions affect forensic traces?
4. Can we detect tampering in HEIF images using block-based analysis?



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# Others Team Members



Prof Sebastiano Battiato

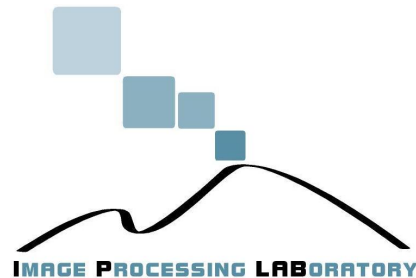


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